

**Suitability Modeling of Sites for the Development of
Floating Offshore Wind Turbines off the Oregon Coast**

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Introduction

On September 3rd, 2024, the Bureau of Ocean Energy Management (BOEM) posted a Final Sale Notice for an offshore wind lease for sites off the Oregon Coast. An auction was scheduled for October 15th, 2024, and five companies were qualified to bid (Bureau of Ocean Energy Management [BOEM], 2024). However, the Confederated tribes of Coos, Lower Umpqua, and Siuslaw sued the BOEM, seeking a delay in the auction until there could be a review of environmental and cultural impacts. While the tribes did express skepticism over floating offshore wind development, Rick Eichstaedt, an attorney for the tribes, stated that they are not entirely opposed to the development of floating offshore wind. They were asking for the BOEM to conduct a more thorough review, as they were left without answers when they called for a further review of the environmental and cultural impacts. They wanted answers related to impacts and a plan for mitigation as early as possible, before the leases were approved (Samayoa, 2024a). By September 25th, 2024, three companies that were set to bid had withdrawn from the bidding process, as opposition to the project mounted from residents and tribes (Samayoa, 2024b). Oregon Governor Tina Kotek expressed concerns as well. Finally, on September 27th, 2024, BOEM completely cancelled its scheduled October auction (Samayoa, 2024c).

While these barriers to development may seem inevitable, they can be avoided by using spatial analysis methods, such as suitability models, to identify sites that are suitable for the development of floating offshore wind turbines. This project involves using a suitability model to identify such sites off the Oregon Coast to avoid repeating the mistakes that led to the BOEM cancelling its scheduled auction for an offshore wind lease. The research question this project seeks to answer is: Using a suitability model, what sites are the most suitable for floating offshore wind development? By identifying sites that avoid ecologically sensitive areas and unsuitable bathymetry, this project aims to answer that question with a two-phase suitability model.

While there is emerging research that exhibits positive impacts of offshore wind farms increasing the biodiversity of an area, the overall impacts remain unclear, according to a case study focused on floating offshore wind in Northern European seas (Lloret, 2025). With the impacts unclear and the concerns raised by various stakeholders in Oregon, Marine Protected Areas ought to be excluded from the sites suitable for floating offshore wind development. There is also an observed “edge effect” near Marine Protected Areas of approximately 1 kilometer (Ohayon, et. al., 2021). This should form another criteria of sites that are considered suitable for floating offshore wind development being at least 1 km away from a Marine Protected Area.

Seafloor depth is also a crucial component of a site’s suitability for floating offshore wind development. The minimum seafloor depth that is suitable for floating offshore wind development is 60 meters (Oregon Department of Energy, 2022). The exact maximum suitable depth is changing with new and emerging technology, but the Hywind Tampen project off the coast of Norway has a seafloor depth of 267-310 meters (Frossard, et. al., 2025). This should form a criteria of sites that are considered suitable for offshore wind development having a seafloor depth between 60 meters and 300 meters.

There are migratory sea bird species in Oregon that are classified as threatened under the Endangered Species Act. One species is the Marbled Murrelet, which is classified as threatened (US Fish & Wildlife Service, 2024). Marbled Murrelets also tend to forage near-shore, specifically within a few miles of shore and near tides and river currents (Cornell Lab of Ornithology, n.d.-a). With “a few” being as low as three, a potential buffer zone around critical habitats could be rounded up to 5 km to provide more defensible space in a suitability model.

While the Brown Pelican was removed from the Federal List of Endangered and Threatened Wildlife in 2009 (US Fish & Wildlife Service, 2009), it is still a species protected by the Migratory Bird Treaty Act. Brown Pelicans also tend to forage within 12 miles of shore (Cornell Lab of Ornithology, n.d.-b). By converting 12 miles into kilometers and rounding upwards for a more conservative suitability model, a potential buffer zone around Brown Pelican Roosts could be defined by a radius of 20 km.

In addition to seafloor depth, there are other bathymetric characteristics that are relevant to floating offshore wind development. One relevant characteristic is slope. Another suitability model pertaining to the US Mid-Atlantic Coast classified sites with a slope that is less than 10 degrees as suitable for floating offshore wind turbines using standard technology. However, there has been development of floating offshore wind on steeper slopes with more advanced technology (Rodenbach, 2024). This will be factored into the methodology of the study as well, potentially using a scoring system as part of the second phase of the suitability model.

Methodology

This suitability model initially involved manually completing each step in *ArcGIS Pro* before replicating it with the ModelBuilder tool. The built-in Suitability Modeler was not used for this project. The original data included a layer for NOAA’s MPA Inventory (National Oceanic and Atmospheric Administration, 2023), a layer for Brown Pelican Survey Roosts (National Audubon Society, n.d.), a layer for Marbled Murrelet Critical Habitats (US Fish & Wildlife Service, 2020), and a Digital Elevation Model called TopoBathy (Esri, 2014). A model/workflow diagram is linked in Figs. 1a-1c, and a metadata table is linked in Fig. 2.

Figure 1a: Model/workflow Diagram from ModelBuilder (full graphic)

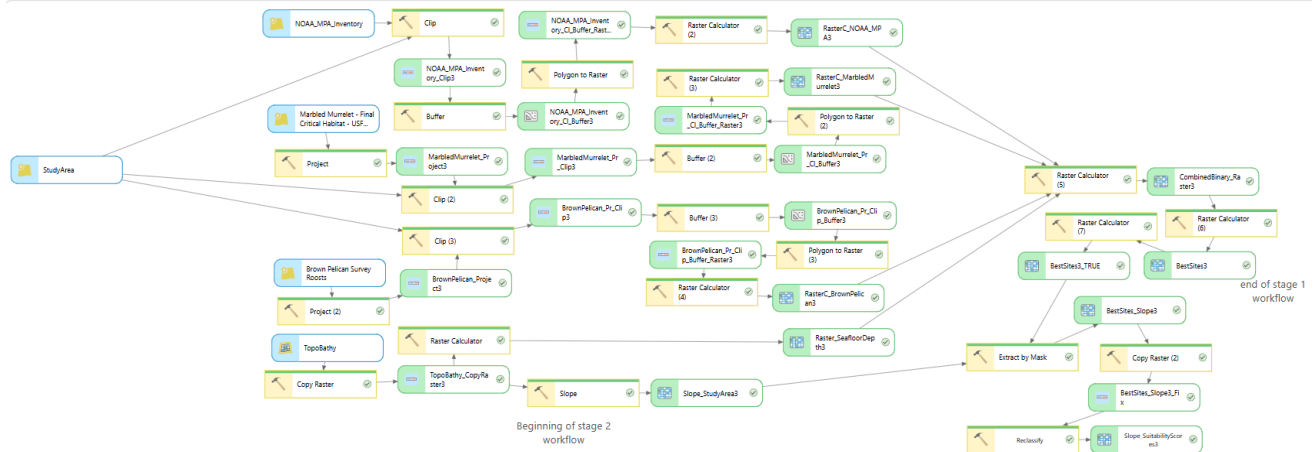


Fig. 1a: Model/workflow diagram made using the ModelBuilder tool in ArcGIS Pro. This is the full graphic. Enlarged portions of this graphic are found in Figs. 1b and 1c.

Figure 1b: Enlarged portion of Model/Workflow Diagram (Left side)

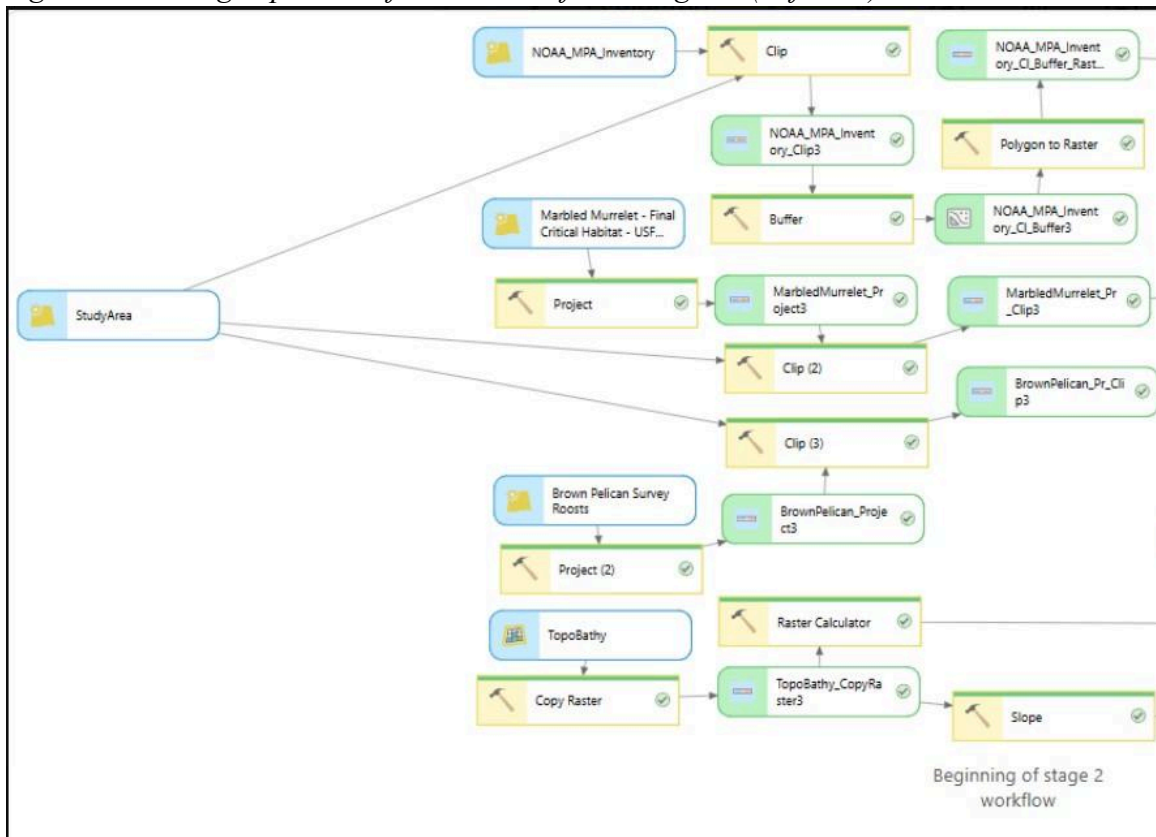


Fig. 1b: Enlarged portion of workflow diagram showing the left side of the diagram. Part of the second stage as well as part of the first stage is shown in the diagram. The right side portion of the enlarged diagram includes the end of the first phase and proceeds where the beginning of the second phase of the workflow was cut off.

Fig. 1c: Enlarged portion of the model/workflow diagram (right side)

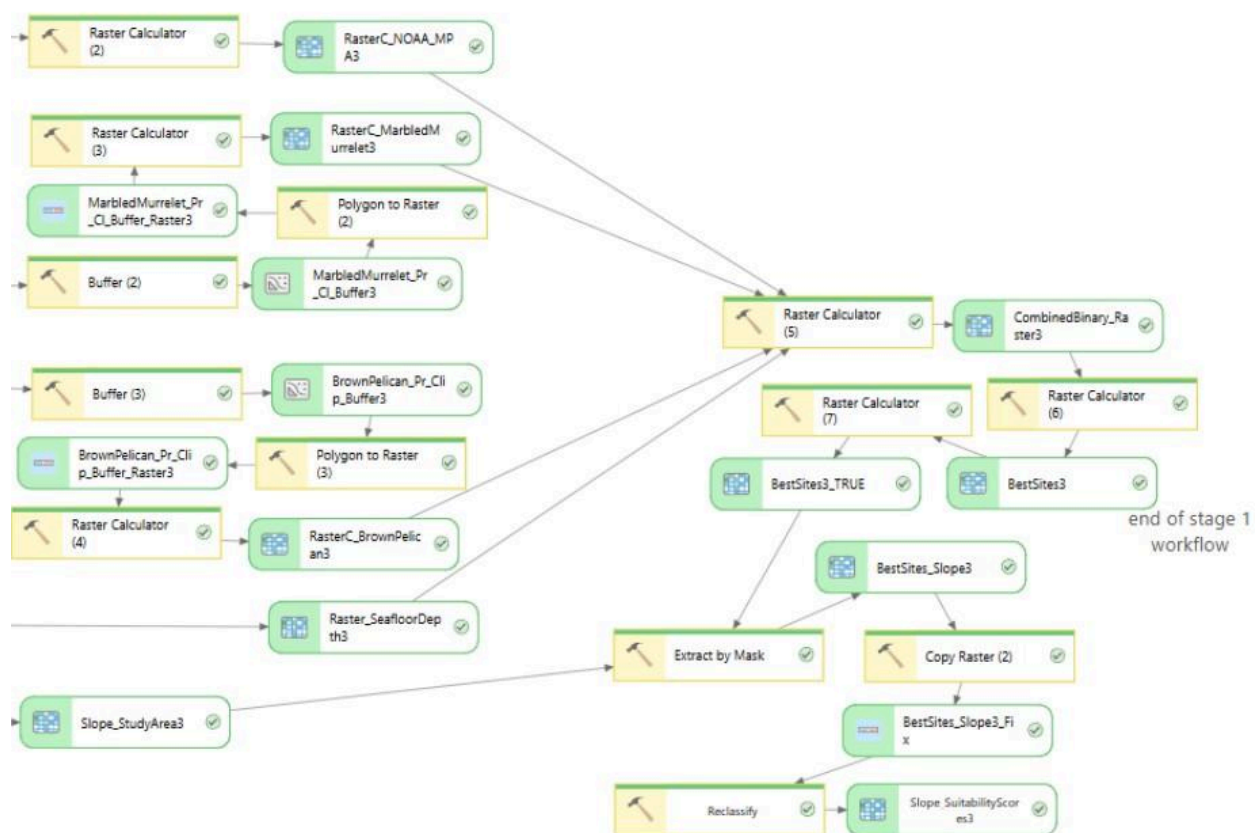


Fig. 1c: This other portion of the enlarged diagram shows the final steps of the process.

Figure 2: Metadata table

Layer Name	Date created	Original scale/resolution	PCS	GCS (datum)	Creator of data (include URL of the download page)	Useful Attributes	Data Quality Issues	Method of creation	Relevance to analysis
Brown Pelican Roosts Updated	2016-03-03	0.0001 meters	None	WGS 1984	Audubon Society- with data from Ebird and USFWS https://www.arcgis.com/home/item.html?id=f1f1f699cad24e1fa61159c1c2a40fbb	Shape Area, Polygon	Needs clipping	ISO 19139	Protected by migratory bird treaty act
NOAA Marine Protected Areas (MPA) Inventory	2023-03-29	0.0001 meters	WGS 1984 Web Mercator (auxiliary sphere)	WGS 1984	NOAA https://www.arcgis.com/home/item.html?id=8663945e885946e09f952d61b0e9652b	Shape Area, No Table	Needs clipping	ISO 19139	Marine Protected Areas are sensitive areas
Topobathy	2025-06-09	0.0001 meters	WGS 1984 Web Mercator (auxiliary sphere)	WGS 1984	Esri https://www.arcgis.com/home/item.html?id=c753e5bfadb54d46b69c3e68922483bc	Ocean floor depth	Raster data hard to use polygons/integrate with them, large raster needs to be clipped and calculated	ISO 19139	Ocean floor depth is important for wind turbines , some depths are optimal
Marbled Murrelet - Final Critical Habitat - USFWS [ds157]	2020-02-11	0.0001 meters	None	WGS 1984	USFWS https://www.arcgis.com/home/item.html?id=3135c4f5b2e2488987f88c380694d9d0	Shape area, Polygon	No exact location in table	ISO 19115	Marbled Murrelets are migratory, so critical habitats are important to not be disturbed

Fig. 2: Table showing the metadata of each data source/layer.

The workflow began by defining a study area along the Oregon Coast, as well as acquiring all the preliminary data as shown in Figs. 1a-2. Two layers, the Brown Pelican Survey Roosts and the Marbled Murrelet Critical Habitats, lacked a projected coordinate system. To ensure clean, matching data and avoid errors, both layers were projected to the “WGS 1984 Web Mercator (auxiliary sphere)” projected coordinate system, using the “Project” tool in *ArcGIS Pro*. This produced two projected layers: 1. MarbledMurrelet_Project3 and 2. BrownPelican_Project3.

With all layers now in a matching projection, all layers other than the TopoBathy layer were clipped to the study area. Using the “Clip” tool in *ArcGIS Pro*, this produced three newly clipped layers: 1. NOAA_MPA_Inventory_Clip3, 2. MarbledMurrelet_Pr_Clip3, and 3. BrownPelican_Pr_Clip3.

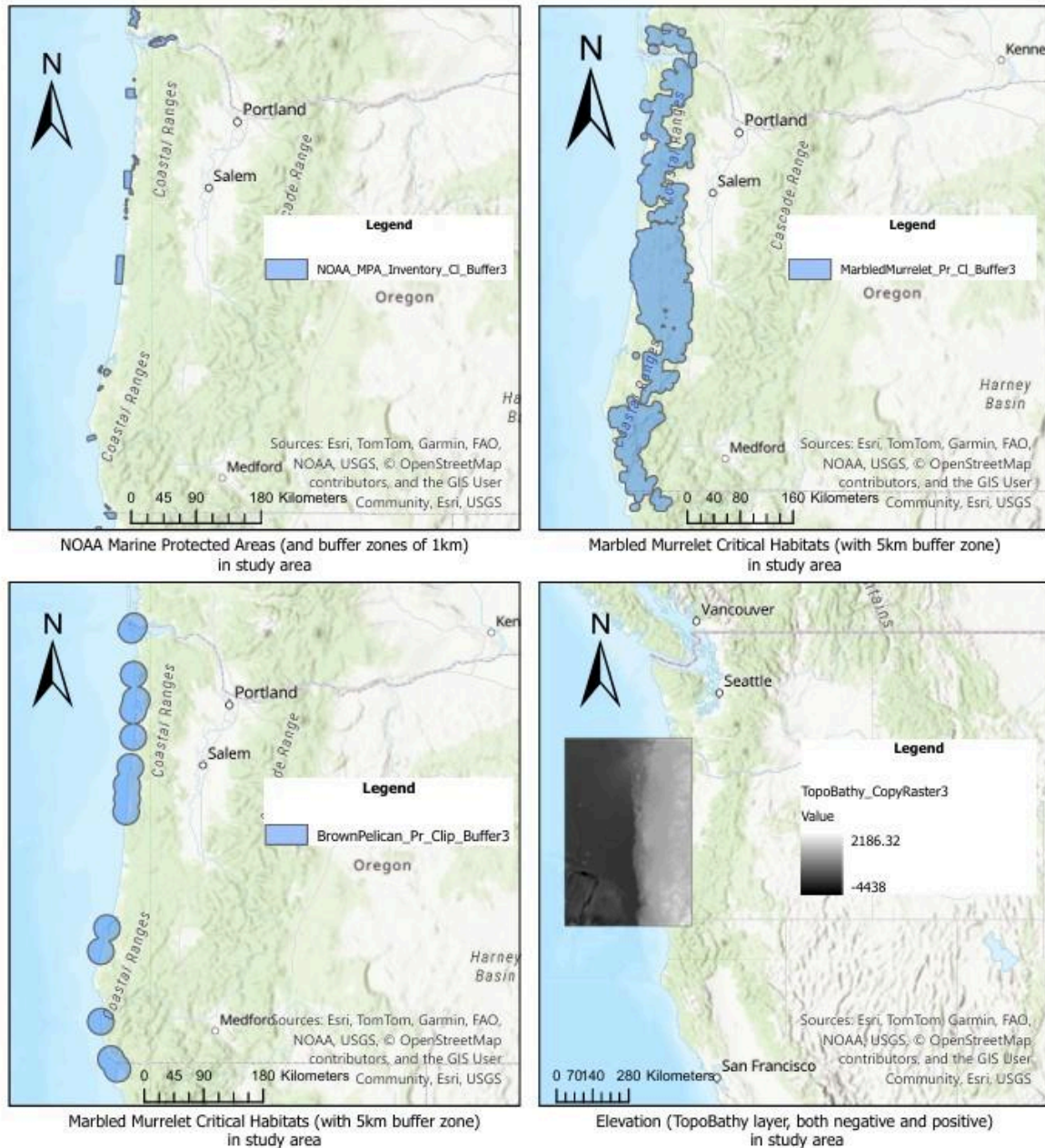
The TopoBathy layer needed to be copied to be usable, as it was such a large raster. A raster the size of the entire earth and with such a small cell size of 0.0001 meters was unable to be efficiently processed. Using the “Copy Raster” tool in *ArcGIS Pro*, it was copied to the extent of the study area and with an adjusted cell size of 500 meters. It used a 32-bit float as well. This produced a new layer: TopoBathy_CopyRaster3.

The clipped polygon layers that excluded TopoBathy_CopyRaster3 were all layers for either protected areas or migratory bird habitats. These layers needed to have defensible space around the polygons. In the case of NOAA_MPA_Inventory_Clip3, it would be given a 1 km buffer around the entirety of each polygon within the layer. In the case of MarbledMurrelet_Pr_Clip3, it would be given a 5 km buffer around the entirety of each polygon within the layer. In the case of BrownPelican_Pr_Clip3, it would be given a 20 km buffer around the entirety of each polygon within the layer. This project used the “Buffer” tool in *ArcGIS Pro* to create the aforementioned “buffer zones.” This produced three new layers that include both the polygons and their relevant buffer zones: 1. NOAA_MPA_Inventory_Cl_Buffer3, 2. MarbledMurrelet_Pr_Cl_Buffer3, and 3. BrownPelican_Pr_Clip_Buffer3. All three layers, along with TopoBathy, are shown in Fig. 3.

Figure 3: Clipped polygons with buffers and copied/clipped TopoBathy raster

Map of criteria included in binary variables

The various criteria below are included in the binary variables. Such criteria are mapped below.



Map created by: Brady Shields. 6/8/26
(Affiliation: Oregon State University CEOAS)

Fig. 3: The layers with criteria pertaining to the binary variables. All polygons have matching projections, are clipped to the study area, and include relevant buffer zones. The TopoBathy Layer is copied to the extent of the study area with a more usable cell size. Map produced using the "Layout" tool in *ArcGIS Pro*.

For the polygon/vector layers to be compatible with the raster layer for analysis, it was necessary to rasterize the data. All three relevant polygon layers were rasterized using the “Polygon to Raster” tool in *ArcGIS Pro*. Using this tool, the raster cell sizes were adjusted to 500 meters. And the rasters used a 32-bit float as well. This tool produced three new layers: 1.

NOAA_MPA_Inventory_Cl_Buffer_Raster3, 2. MarbledMurrelet_Pr_Cl_Buffer_Raster3, and 3. BrownPelican_Pr_Clip_Buffer_Raster3.

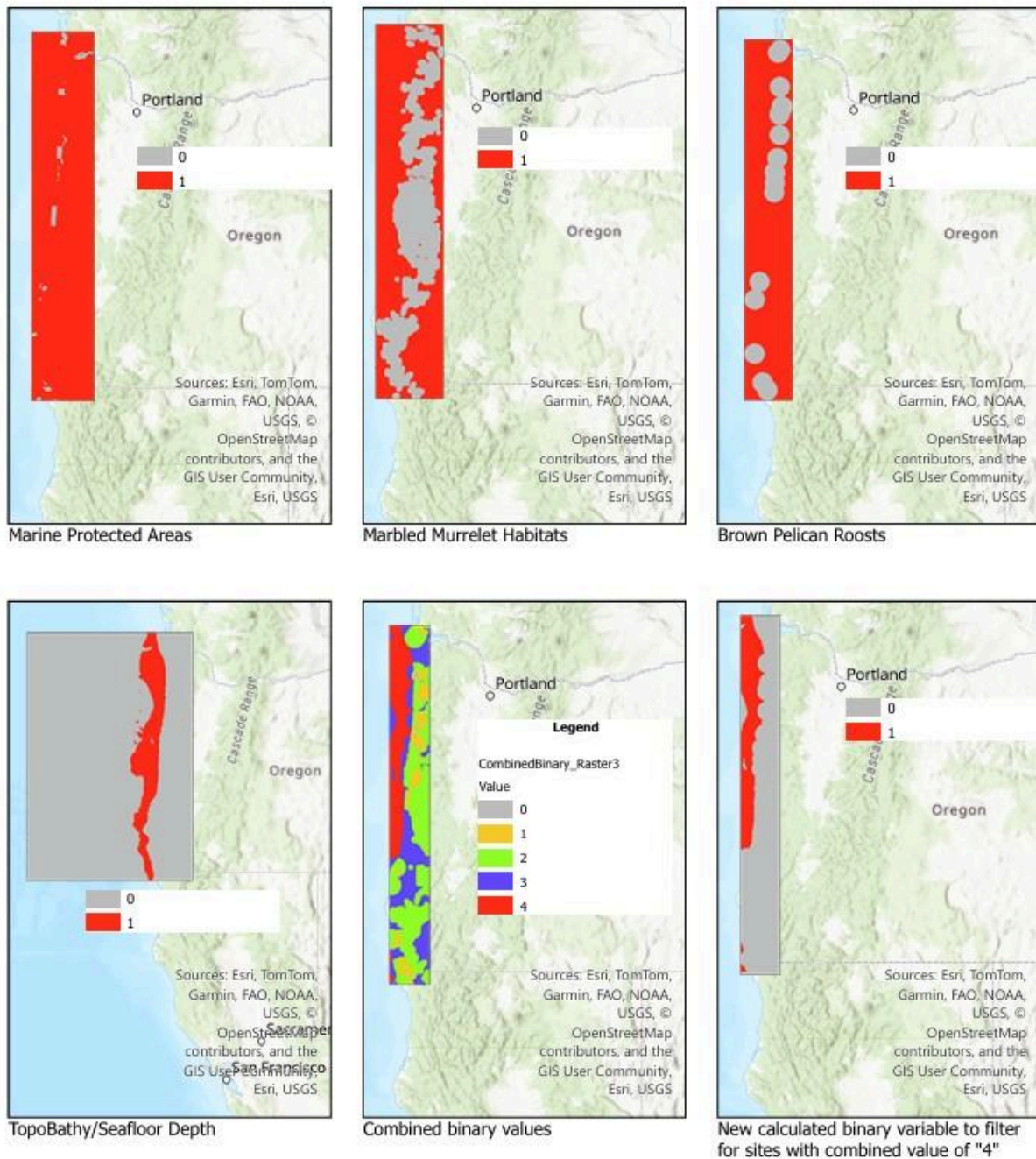
With all of the data rasterized and compatible, it was now time to calculate each raster with binary values. Using the “Raster Calculator” tool in *ArcGIS Pro*, each raster was calculated to have sites with either a value of “0” or “1.” A value of “1” means that the site meets the suitability criteria for that specific variable. A value of “0” means that the site does not meet the suitability criteria for that specific variable. This tool produced a total of four new layers: 1. RasterC_NOAA_MPA3, 2. RasterC_MarbledMurrelet3, 3. RasterC_BrownPelican3, and 4. RasterC_SeafloorDepth3. RasterC_NOAA_MPA3 classified the binary values on the basis of whether the site was within the rasters in NOAA_MPA_Inventory_Cl_Buffer_Raster3, with sites in the rasters being assigned a value of 0, and sites outside the rasters being assigned a value of 1. RasterC_MarbledMurrelet3 classified the binary values on the basis of whether the site was within the rasters in MarbledMurrelet_Pr_Cl_Buffer_Raster3, with sites in the rasters being assigned a value of 0, and sites outside the rasters being assigned a value of 1. RasterC_BrownPelican3 classified the binary values on the basis of whether the site was within the rasters in BrownPelican_Pr_Clip_Buffer_Raster3, with sites in the rasters being assigned a value of 0, and sites outside the rasters being assigned a value of 1. RasterC_SeafloorDepth3 was classified based on the site’s seafloor depth according to TopoBathy_CopyRaster3, with sites with an elevation value greater than or equal to -300 meters and less than or equal to -60 meters being assigned a value of 1, and sites outside that range being assigned a value of 0.

The “Raster Calculator” tool was used again in *ArcGIS Pro*, in order to determine combined binary values of each calculated raster. The values of each site according to RasterC_NOAA_MPA3, RasterC_MarbledMurrelet3, RasterC_BrownPelican3, and RasterC_SeafloorDepth3 were all added together in the “Raster Calculator” tool in *ArcGIS Pro*. This produced a new layer: CombinedBinary_Raster3, which had values ranging from 0-4. Sites with a value of 4 were the only sites that were considered suitable for floating offshore wind development. After the combined binary values were assigned on a scale of 0-4, the “Raster Calculator” tool was used again in *ArcGIS Pro*, and this time it was used to calculate a new binary raster with values of “0” and “1” to determine the best sites that met the criteria tests of all four binary variables. Sites with a value of 4 according to CombinedBinary_Raster3 were now assigned a value of 1, while sites with value of anything other than 4 in CombinedBinary_Raster3 were now assigned a value of 0. This produced a new layer called BestSites3. Maps depicting the values for each site according to RasterC_NOAA_MPA3, RasterC_MarbledMurrelet3, RasterC_BrownPelican3, RasterC_SeafloorDepth3, CombinedBinary_Raster3, and BestSites3 are found in Fig. 4.

Figure 4: Maps depicting every site's binary value

Calculated Binary Variables (Raster Calculator)

Each variable has been rasterized (if not already) and every spot within the study area has been assigned a value of 0 (unsuitable) or 1 (suitable) based on variable criteria. However, the bottom middle map is the combined value of each raster/variable from 0-4. Only 4 is considered suitable, so the bottom right map is the final binary map for which sites are suitable in the first phase of the workflow.



Map created by: Brady Shields. 6/8/26
(Affiliation: Oregon State University CEOAS)

Fig. 4: Each map, other than the Combined binary values map, has sites with a value of either 0 or 1. These variables represent the end of the first stage of the workflow. Map produced using the "Layout" tool in *ArcGIS Pro*.

After finishing the first phase of the workflow, the next phase focused on identifying the suitability of the sites that passed all four binary variable tests, based on the slope of each site. This new phase of the workflow began by using the “Slope” tool in *ArcGIS Pro*. This tool calculated slope in degrees with TopoBathy_CopyRaster3 as the input raster. This produced a new layer: Slope_StudyArea3. However, there was still more to be done to prepare BestSites3 for this new phase of the workflow. The “Raster Calculator” tool was used again in *ArcGIS Pro* by creating a new raster that only included the sites that had a value of 1 according to BestSites3. This produced a new layer: BestSites3_TRUE.

The “Extract by Mask” tool was used in *ArcGIS Pro*, with BestSites3_TRUE and Slope_StudyArea3 as the inputs. This resulted in a slope raster that was limited to the sites in BestSites3_TRUE. This tool produced a new layer: BestSites_Slope3. However, this layer also needed to be copied as a form of debugging to prepare it for classification. The “Copy Raster” tool was used in *ArcGIS Pro* with BestSites_Slope3 as the input raster. This tool produced a new layer: BestSites_Slope3_Fix.

With a clean, debugged layer displaying the slope of each site that passed the four binary tests of the first phase of the workflow, it was now time to create a suitability scoring system based on the slope values in degrees. This involved using the “Reclassify” tool in *ArcGIS Pro*, where the values were classified using a manual interval and 10 classes of values. The maximum slope for all sites was approximately 14.9 degrees. Sites with minimal slopes $< 1^\circ$ were assigned a score of 10. Sites with slopes $\geq 1^\circ$ and $< 4^\circ$ were assigned a score of 9. Sites with slopes $\geq 4^\circ$ and $< 7^\circ$ were assigned a score of 8. Sites with slopes $\geq 7^\circ$ and $< 10^\circ$ were assigned a score of 7. Sites with slopes $\geq 10^\circ$ and $< 10.817^\circ$ were assigned a score of 6. Sites with slopes $\geq 10.817^\circ$ and $< 11.633^\circ$ were assigned a score of 5. Sites with slopes $\geq 11.633^\circ$ and $< 12.45^\circ$ were assigned a score of 4. Sites with slopes $\geq 12.45^\circ$ and $< 13.267^\circ$ were assigned a score of 3. Sites with slopes $\geq 13.267^\circ$ and $< 14.083^\circ$ were assigned a score of 2. Sites with slopes $\geq 14.083^\circ$ were assigned a score of 1. Under this scoring system, a score of 7 was considered a “passing” score where the slope is suitable for standard floating offshore wind turbine technology. Sites with scores of 1-6 are considered to be less than ideal for floating offshore wind development but still have potential for development under new and emerging technology. This process produced a new layer: Slope_SuitabilityScores3. These values are shown in a table in Fig. 5 and in a final suitability scoring map in Fig. 6. This concludes the second phase of the workflow.

Figure 5: Table of slope values and suitability scores

Slope	Score
$< 1^{\circ}$	10
$\geq 1^{\circ}$ and $< 4^{\circ}$	9
$\geq 4^{\circ}$ and $< 7^{\circ}$	8
$\geq 7^{\circ}$ and $< 10^{\circ}$	7
$\geq 10^{\circ}$ and $< 10.817^{\circ}$	6
$\geq 10.817^{\circ}$ and $< 11.633^{\circ}$	5
$\geq 11.633^{\circ}$ and $< 12.45^{\circ}$	4
$\geq 12.45^{\circ}$ and $< 13.267^{\circ}$	3
$\geq 13.267^{\circ}$ and $< 14.083^{\circ}$	2
$\geq 14.083^{\circ}$	1

Fig. 5: Table shows slope values in degrees and their corresponding suitability score.

Results

The final result of the workflow is a suitability scoring map for each site that passed all four binary tests of the first phase of the workflow. The four binary tests were: 1. Not within 1 kilometer of a Marine Protected Area, 2. Not within 5 kilometers of a Marbled Murrelet Critical Habitat, 3. Not within 20 kilometers of a Brown Pelican Roost, and 4. Within the defined suitable seafloor depth range of 60-300 meters. Any site that passed all four binary tests was included in the sample for the second phase of the workflow, and in the second phase, all sites were scored according to their slope on a scale of 1-10 with 10 being the most suitable and 1 being the least suitable. Sites with a score of 7 are meeting the minimum score to be considered suitable for development in the immediate future, while sites with a score of 1-6 are difficult to develop at the moment, although they do have some degree of potential. However, such potential is dependent on technological advancements and whether energy markets can justify a higher cost. Fig. 6 displays the final results of the second phase of the workflow in the form of a suitability scoring system.

Figure 6: Suitability score map of all sites that passed the binary tests of the first phase of the workflow.

Suitability Scores for floating offshore wind sites off the Oregon Coast

A map showing the suitability of sites for floating offshore wind turbines off the Oregon Coast. Sites included in the scoring system passed a total of FOUR (4) binary tests: 1. Seafloor depth range of 60-300m. 2. Not in a Marine Protected Area. 3. Not within 5km of a Marbled Murrelet Critical Habitat, 4. Not within 20km of a Brown Pelican Roost.

The scoring system is based on slope with all sites in the final sample having roughly 14.9° slope or less. Sites with $\leq 1^\circ$ receive a score of 10. Sites with $>1^\circ$ and $\leq 4^\circ$ receive a score of 9. Sites with $>4^\circ$ and $\leq 7^\circ$ receive a score of 8. Sites with $>7^\circ$ and $\leq 10^\circ$ receive a score of 7. For sites with slopes $>10^\circ$, they are scored in intervals of roughly 0.8167° .

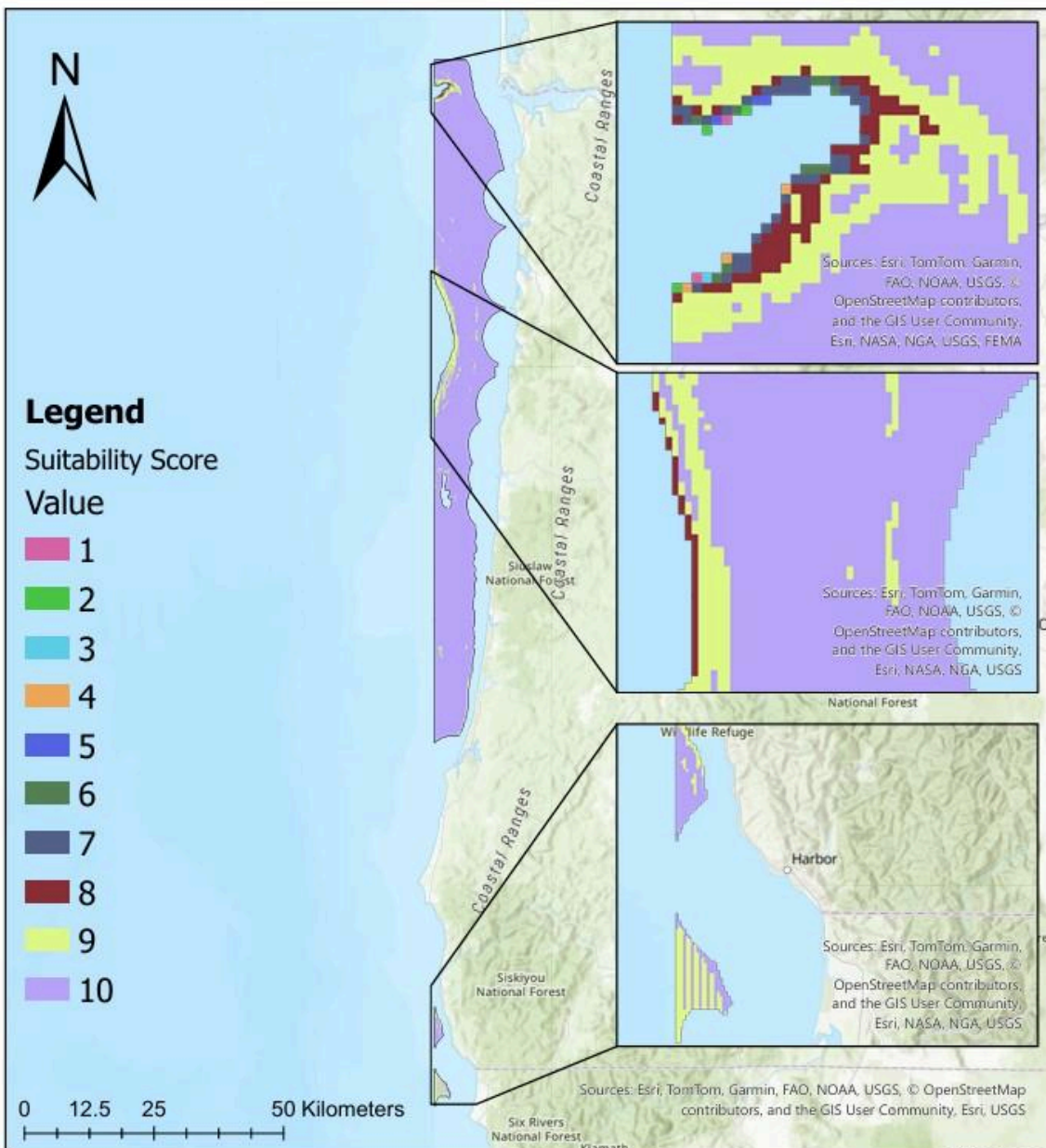


Fig. 6: This map shows suitability scores of 1-10 for sites that have passed all four binary tests in the first phase of the workflow.

The map in Fig. 6 displays large portions of the ocean off the Oregon Coast with a suitability score of 10, therefore showing much of it being suitable for floating offshore wind development. The inset maps show many sites with a value of 9 along the Southern Coast, Northern Coast, and North-Central Coast. The Northern and North-Central Coast Areas also have sites with values of 8, and the Northern Coast shows many sites with values of 7 or less. This shows large portions of the Oregon Coast being considered suitable for floating offshore wind development.

Discussion

The initial research question that this project sought to answer was: Using a suitability model, what sites are the most suitable for floating offshore wind development?

The suitability scoring model produced by this project suggests that the Central Coast has the best sites for floating offshore wind development, due to it having substantial and uninterrupted swaths of ocean floor with a suitability score of 10. The criteria of the model that is necessary to produce a suitability score of 10 indicates that those sites are not within 1 kilometer of a Marine Protected Area, not within 5 kilometers of a Marbled Murrelet Critical Habitat, not within 20 kilometers of a Brown Pelican Survey Roost, within a seafloor depth range of 60 to 300 meters deep, and have a slope of $< 1^\circ$.

At the same time, there are also some sites within the Central Coast region that have scores of 9 and 8. This indicates all the same criteria as the sites with scores of 10, aside from slope. Sites with scores of 9 have a slope $\geq 1^\circ$ and $< 4^\circ$, and sites with scores of 8 have a slope $\geq 4^\circ$ and $< 7^\circ$. All of these sites are easily suitable for floating offshore wind development under such criteria.

This model also produced very few suitable sites for floating offshore wind development off the Southern Oregon Coast. As previously mentioned in this report, the auction that caused the most controversy was the auction for sites off the Southern Oregon Coast. This suitability model therefore validates the concerns from tribes who were not entirely opposed to floating offshore wind development, but were instead concerned about the potential environmental impact.

By avoiding sites within 1 kilometer of Marine Protected Areas, 5 kilometers of Marbled Murrelet Critical Habitats, and 20 kilometers of Brown Pelican Roosts, this model takes the concerns of environmental impact from tribes very seriously by excluding those sites from floating offshore wind development entirely.

The model also factors in concerns potentially raised by engineers and developers regarding the bathymetric profile of potential sites for floating offshore wind development. These include seafloor depth and slope. By excluding sites with a seafloor depth outside the suitable range of 60-300 meters from floating offshore wind development entirely, this model takes the concerns about physical suitability seriously as well. By scoring the sample of sites that passed the binary tests based on their slope, we can identify the sites that are possible to develop from least difficult to most difficult.

By factoring in concerns of various stakeholders and developing a two-phase workflow model that tests multiple binary variables in the first phase and scores sites that passed the binary tests based on a variable with a range of values in the second phase, this model successfully identifies sites that meet the concerns of environmentalists, tribes, engineers, and developers.

However, this model is not without its limitations. By adjusting the cell size to 500 meters from the original 0.0001 meters, this study was highly generalized and far less detailed than the resolution of the original spatial data. This could have led to many details being omitted through generalization. The rasters could be analyzed with a cell size of 10 meters and in increments of different sections of the coast. If this model were to be replicated by taking the study area, placing a grid with 50 cells of equal area over the study area, and analyzing the rasters with an adjusted cell size of 10 meters instead of 500 meters, then such a model may produce very different results.

The model and workflow also omits other factors, such as shipping corridors, submarine cables, wind speeds, fisheries, kelp canopy cover, and distance from the shore. There were attempts to add other layers, such as wind speeds, kelp canopy cover, and distance from the shore, but they were either unavailable or too difficult to analyze. If these factors could be included in the model, it may show more nuance in the aforementioned “substantial and uninterrupted swaths of ocean floor with a suitability score of 10.”

In the future, this model and similar models could analyze much smaller areas, adjust the cell size to 10 meters instead of 500 meters to reduce the simplification of generalization, and include other variables such as shipping corridors, submarine cables, wind speeds, fisheries, kelp canopy cover, and distance from the shore. These variables factor in additional concerns from various stakeholders, including environmentalists, tribes, fishing sector, engineers, developers, shipping companies, seafarers, telecommunications providers, etc.

Addressing these concerns and developing suitability models is important for the transition to renewable energy because avoiding lawsuits and liability can help expedite the process and avoid permanent bureaucratic “red tape.”

Overall, this project used a model that excluded the most ecologically sensitive sites and the least geologically suitable sites, and the resulting output produced many sites that are highly suitable and sit far away from the site of the original controversial auction for floating offshore wind development off the Oregon Coast. This model effectively addresses the concerns of stakeholders, such as environmentalists, tribes, engineers, and developers. However, it has its limitations due to the spatial resolution of the data and the required generalization to analyze such a large area. It also has limitations due to a lack of additional variables that could show more nuance in the general area that groups sites deemed generally suitable for floating offshore

wind development. Addressing these concerns from stakeholders, including the aforementioned environmentalists, tribes, engineers, and developers, as well as the fishing sector, shipping companies, seafarers, telecommunications providers, etc. is important for effective offshore wind development because it avoids lawsuits and liability issues for everyone involved and helps expedite the process of the transition to renewable energy.

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